

## AKYAR, Russian Federation

WMO: 350370

Lat: 51.873N

Lon: 58.223E

Elev: 1119

StdP: 14.11

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -20.8      | -15.5 | -26.6                         | 1.3 | -20.6 | -21.4 | 1.8 | -15.2 | 24.7                  | -4.6 | 21.9 | 3.6  | 6.3                   | 300  | 0.630 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 20.6                   | 90.3            | 64.4 | 86.7 | 63.3 | 83.6 | 62.4 | 67.2                | 82.4 | 65.7 | 81.0 | 64.2 | 79.0 | 8.4                  | 230  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 62.6                            | 88.3 | 71.9 | 60.7 | 82.5 | 70.7 | 58.7 | 76.9 | 69.6 | 32.3          | 82.0 | 31.1 | 81.0 | 29.9 | 78.9 | 76.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |     |
|-------------------|------|------|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|-----|
|                   |      |      | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |     |
| 23.6              | 20.7 | 18.1 | DB                         | -25.2 | 94.0               | 5.8 | 3.1                                                | -29.4 | 96.2       | -32.8 | 97.9       | -36.1 | 99.7       | -40.4 | 101.9 | (4) |
|                   |      |      | WB                         | -25.4 | 69.7               | 5.5 | 2.7                                                | -29.4 | 71.7       | -32.6 | 73.3       | -35.7 | 74.8       | -39.8 | 76.8  | (5) |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 38.8  | 6.1   | 8.3   | 20.1  | 41.2  | 57.1  | 65.6  | 68.2  | 66.4  | 54.6  | 40.2  | 24.3  | 11.9  | (6)  |
| (7)  |                                                                   | DBStd       | 24.76 | 11.96 | 11.79 | 9.78  | 11.38 | 8.16  | 7.68  | 6.43  | 7.95  | 8.02  | 9.33  | 10.83 | 11.97 | (7)  |
| (8)  |                                                                   | HDD50       | 6110  | 1360  | 1168  | 926   | 308   | 28    | 2     | 0     | 1     | 41    | 323   | 771   | 1182  | (8)  |
| (9)  |                                                                   | HDD65       | 9951  | 1825  | 1588  | 1391  | 713   | 268   | 85    | 40    | 82    | 323   | 768   | 1221  | 1647  | (9)  |
| (10) |                                                                   | CDD50       | 2037  | 0     | 0     | 0     | 45    | 247   | 472   | 565   | 509   | 180   | 19    | 0     | 0     | (10) |
| (11) |                                                                   | CDD65       | 403   | 0     | 0     | 0     | 0     | 22    | 104   | 140   | 125   | 12    | 0     | 0     | 0     | (11) |
| (12) |                                                                   | CDH74       | 4611  | 0     | 0     | 0     | 13    | 344   | 1202  | 1510  | 1378  | 162   | 2     | 0     | 0     | (12) |
| (13) |                                                                   | CDH80       | 1594  | 0     | 0     | 0     | 1     | 60    | 432   | 520   | 532   | 49    | 0     | 0     | 0     | (13) |
| (14) | Wind                                                              | WSAvg       | 8.1   | 7.7   | 8.5   | 8.8   | 9.1   | 8.5   | 8.0   | 8.1   | 7.4   | 7.2   | 7.9   | 8.4   | 8.0   | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 14.3  | 1.1   | 0.7   | 1.1   | 1.1   | 1.3   | 1.7   | 1.9   | 1.5   | 0.9   | 1.0   | 1.0   | 1.1   | (15) |
| (16) |                                                                   | PrecMax     | 24.6  | 2.1   | 2.4   | 2.5   | 3.1   | 2.5   | 4.4   | 4.9   | 5.9   | 2.1   | 1.9   | 1.7   | 2.4   | (16) |
| (17) |                                                                   | PrecMin     | 8.9   | 0.2   | 0.1   | 0.0   | 0.1   | 0.2   | 0.7   | 0.4   | 0.2   | 0.0   | 0.0   | 0.0   | 0.3   | (17) |
| (18) |                                                                   | PrecStd     | 4.1   | 0.5   | 0.5   | 0.7   | 0.8   | 0.7   | 0.9   | 1.2   | 1.3   | 0.6   | 0.5   | 0.5   | 0.6   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | DB          | 32.5  | 34.1  | 40.8  | 76.3  | 84.6  | 93.3  | 94.8  | 94.2  | 86.3  | 71.4  | 48.5  | 34.6  | (19)  |      |
| (20) |                                                                   | MCWB        | 31.4  | 32.1  | 35.7  | 54.9  | 59.2  | 64.6  | 64.3  | 64.4  | 60.4  | 53.4  | 43.0  | 33.2  | (20)  |      |
| (21) |                                                                   | DB          | 29.7  | 31.3  | 37.1  | 70.7  | 81.5  | 88.8  | 89.3  | 89.8  | 78.1  | 64.4  | 43.1  | 32.4  | (21)  |      |
| (22) |                                                                   | MCWB        | 28.5  | 29.9  | 33.9  | 52.8  | 58.3  | 64.2  | 65.0  | 64.0  | 56.8  | 50.6  | 40.0  | 31.4  | (22)  |      |
| (23) |                                                                   | DB          | 27.4  | 29.0  | 34.9  | 64.8  | 78.1  | 85.3  | 85.4  | 86.0  | 73.5  | 59.4  | 39.6  | 30.5  | (23)  |      |
| (24) |                                                                   | MCWB        | 26.2  | 27.8  | 32.7  | 50.0  | 57.8  | 63.2  | 63.6  | 63.4  | 55.9  | 46.9  | 36.8  | 29.3  | (24)  |      |
| (25) |                                                                   | DB          | 24.2  | 25.6  | 33.2  | 59.0  | 73.7  | 81.3  | 82.3  | 82.0  | 68.8  | 54.8  | 36.9  | 27.9  | (25)  |      |
| (26) |                                                                   | MCWB        | 23.0  | 24.3  | 31.5  | 47.6  | 56.2  | 62.0  | 63.1  | 61.9  | 53.9  | 45.5  | 34.8  | 26.6  | (26)  |      |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | WB          | 31.7  | 32.6  | 36.9  | 57.3  | 62.4  | 68.9  | 69.6  | 68.9  | 62.6  | 54.7  | 44.8  | 33.4  | (27)  |      |
| (28) |                                                                   | MCDB        | 32.2  | 33.5  | 39.1  | 73.1  | 77.5  | 84.3  | 84.0  | 84.1  | 82.2  | 67.2  | 46.1  | 34.3  | (28)  |      |
| (29) |                                                                   | WB          | 28.7  | 30.1  | 34.6  | 53.7  | 60.8  | 66.5  | 67.6  | 66.3  | 59.5  | 51.9  | 40.2  | 31.4  | (29)  |      |
| (30) |                                                                   | MCDB        | 29.6  | 31.1  | 36.5  | 66.9  | 76.4  | 83.1  | 81.8  | 83.2  | 73.1  | 61.6  | 42.7  | 32.1  | (30)  |      |
| (31) |                                                                   | WB          | 26.2  | 27.7  | 33.1  | 51.2  | 59.2  | 64.8  | 65.9  | 64.7  | 57.2  | 48.8  | 37.1  | 29.4  | (31)  |      |
| (32) |                                                                   | MCDB        | 27.3  | 28.9  | 34.7  | 62.7  | 74.3  | 80.9  | 79.6  | 81.1  | 69.0  | 56.9  | 39.4  | 30.4  | (32)  |      |
| (33) |                                                                   | WB          | 23.1  | 24.3  | 31.7  | 48.3  | 57.4  | 63.2  | 64.4  | 63.2  | 54.9  | 46.2  | 34.9  | 26.7  | (33)  |      |
| (34) |                                                                   | MCDB        | 24.2  | 25.5  | 33.2  | 58.1  | 71.8  | 77.9  | 77.5  | 78.1  | 66.8  | 52.7  | 36.8  | 27.8  | (34)  |      |
| (35) | Mean Daily Temperature Range                                      | MDBR        | 12.3  | 13.8  | 13.9  | 17.5  | 21.9  | 21.7  | 20.6  | 21.6  | 19.9  | 15.9  | 10.1  | 11.6  | (35)  |      |
| (36) |                                                                   | 5% DB       | 10.8  | 11.8  | 12.5  | 26.3  | 27.7  | 27.1  | 26.0  | 26.9  | 27.1  | 26.1  | 12.2  | 10.7  | (36)  |      |
| (37) |                                                                   | MCWBR       | 10.2  | 10.8  | 10.4  | 12.2  | 10.9  | 9.6   | 8.7   | 8.8   | 11.6  | 13.8  | 9.7   | 10.5  | (37)  |      |
| (38) |                                                                   | 5% WB       | 10.3  | 12.0  | 11.0  | 24.0  | 24.5  | 23.9  | 21.6  | 23.5  | 23.3  | 22.7  | 11.1  | 10.4  | (38)  |      |
| (39) |                                                                   | MCWBR       | 10.2  | 11.2  | 9.6   | 12.2  | 11.3  | 9.2   | 8.6   | 8.7   | 11.1  | 12.9  | 9.3   | 10.3  | (39)  |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.292 | 0.320 | 0.375 | 0.418 | 0.403 | 0.398 | 0.409 | 0.429 | 0.376 | 0.350 | 0.322 | 0.289 | (40)  |      |
| (41) |                                                                   | taud        | 2.080 | 2.023 | 1.947 | 2.077 | 2.238 | 2.317 | 2.291 | 2.206 | 2.313 | 2.289 | 2.182 | 2.076 | (41)  |      |
| (42) |                                                                   | Ebn at Noon | 215   | 243   | 251   | 258   | 269   | 272   | 266   | 252   | 254   | 234   | 203   | 191   | (42)  |      |
| (43) |                                                                   | Edh at Noon | 25    | 35    | 46    | 46    | 41    | 39    | 39    | 40    | 32    | 27    | 23    | 21    | (43)  |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 389   | 747   | 1199  | 1485  | 1831  | 2025  | 1914  | 1615  | 1140  | 617   | 286   | 250   | (44)  |      |
| (45) |                                                                   | RadStd      | 38    | 49    | 123   | 145   | 159   | 150   | 115   | 124   | 115   | 105   | 38    | 43    | (45)  |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |      |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |      |
| (46) | Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   | (46) |
|      |                        | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |      |
| (47) | Regional (6 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | +189  | +151  | (47) |

Nomenclature: See separate page